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Hypothesis

Population or Universe: An aggregate of objects under study is called a population or universe.

- A universe containing a finite number of individuals or members is called a finite universe.

E.g. The universe of the weights of students in a particular class.

- A universe with infinite number of members is known as an infinite universe.

E.g. The universe of pressures at various points in the atmosphere.

- The universe of concrete objects is an existent universe.
- The collection of all possible ways in which a specified event can happen is called a hypothetical universe.

- A finite subset of a universe is called a sample. A sample is thus a small portion of the universe.
- The number of individuals in a sample is called the sample size.
- The process of selecting a sample from the universe is called sampling.

Note: The theory of sampling is a study of relationship existing b/w a population and samples drawn from the population.

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Parameters of Statistics

The statistical constants of the population such as a mean and variance, etc ~~and so on~~ are known as the parameters.

The statistical concepts of the sample from the members of the sample to estimate the parameters of the population from which the sample has been drawn is known as statistics.

The population mean and variance are denoted by 'μ' and 'σ^2', while those of the samples are given by '~~$\bar{x}$~~ $\bar{x}$' and 's^2' respectively.

Standard Error: The standard deviation of the sampling distribution of a statistic is known as the standard error (S.E.). It plays an important role in the theory of large ~~number of a~~ samples and It forms a basis of ~~a~~ the testing of hypothesis

If t is any statistic, for large sample ~~Then~~ ~~If~~ $z = \frac{t - E(t)}{S \cdot E(t)}$ is normally distributed with mean 0 and S.D. unity

$S \cdot E(t) \longrightarrow$ standard error

We know, $z = \frac{x - \mu}{\sigma}$

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Test of Significance

An important aspect of the sampling theory is to study the test of significance which will ~~to~~ enable us to decide, on the basis of the results of the sample, whether ~~on~~

i) the deviation between the observed sample statistics and the hypothetical parameter value or

ii) the deviation between two sample statistics is significant or might be attributed due to

chance ~~of~~ or the fluctuations of the sampling.

Testing of a statistical hypothesis

Step 1: Null Hypothesis: For applying the tests of ~~so~~ significance, we first set up a hypothesis which is a definite statement about the population parameter called ~~the~~ null hypothesis. Null hypothesis is denoted by H_0.
Null hypothesis is the hypothesis which is tested for possible rejection under the assumptions i.e. it is true. First, we set up H_0 in clear terms.

Step 2: Alternative Hypothesis: Any hypothesis which is complementary to the null hypothesis (H_0) is called an alternative hypothesis. It is denoted by H_1. ~~Ex~~ For eg- If we want to test the null hypothesis, that the population has a specified mean (μ_0). Then, we have $H_0: \mu = \mu_0$

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Then the alternative hypothesis will be :

(i) $H_1 : \mu \neq \mu_0$ (Two tailed alternative hypothesis)

(ii) $H_1 : \mu > \mu_0$ (right tailed hypothesis or ~~one~~ single tailed ~~hyp~~)

(iii) $H_1 : \mu < \mu_0$ (left tailed hypothesis or single tailed)

Hence, alternative hypothesis helps to know whether the test is two tailed or one tailed.

~~Label~~ Level of a significance

The probability of the value of the variate falling in the critical region is known as ~~a~~ level of significance.

Errors in a sampling

The main aim of the sampling is to draw a valid conclusion about the population parameters on the basis of the sample results. In doing this, we may commit the following two types of errors:

I: Type-I error; when H_0 is true, we may reject it.
P(Rej - H_0 when it is true) = P(Reject H_0/H_0)
= α

α is called the size of the type-I error.

II Type-II error: When H_0 is wrong, we may accept it.

P(Accept H_0 when it is wrong) = P(Accept H_0/H_1) = β

where β is the size of type 2 error.

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Note: The values of the test statistic which separates the critical region and acceptance region are called the critical values or significance values. This value depends on (i) the level of significance used (ii) the alternative hypothesis, whether it is one-tailed or two-tailed.

Student's - t-distribution (t - test)

The t-distribution is used when sample size ≤ 30 and the population standard deviation is unknown. t-statistic is defined as

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}}$$

where $S = \cancel{\frac{\sum(x-\bar{x})^2}{n-1}}$ $\sqrt{\frac{\sum(x-\bar{x})^2}{n-1}}$

here, $\bar{x}$ = mean of the sample, μ = population mean, S = standard deviation of population and n is sample size.

Note: If the standard deviation of the sample s is given, then, t-statistic is defined as

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$$\boxed{t = \frac{\bar{x} - \mu}{s\sqrt{n-1}}}$$

Note: The relation between s and S is ~~gi~~

$$ns^2 = (n-1)S^2$$

Ex– A random sample of a size 16 has 53 as a mean. The sum of squares of the deviation from mean is 135. Can this sample be regarded as taken from the population having 56 as a mean? Obtain 95% and 99% confidence limits of the mean of the population.

⟹ ~~Hypothesis~~: Null Hypothesis (Ho): There is no significance difference between the sample mean and hypothetical population mean.

i.e. $\mu = 56$ (t- statistic)

Under Ho, test ~~hypothesis~~ statistic: $t = \frac{\bar{x} - \mu}{S\sqrt{n}}$

Given, $\bar{x} = 53$, $\mu = 56$, $n = 16$,

$\Sigma(x - \bar{x})^2 = 135$

$$S = \sqrt{\frac{135}{15}} = 3$$

$$t = \frac{53 - 56}{3/4} = \frac{-\cancel{3}}{\cancel{3}} = \frac{-1}{1/4} = -4$$

$$|t| = |-4| = 4$$

Conclusion: $\because$ $|t| = 4 > t_{0.05} = 2.13$

i.e. the calculated value of t is more than the tabulated value.

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i.e. the Null Hypothesis is rejected.
Hence, the sample mean has not come from population having 56 as a mean.

99% confidence limit:

95% confidence level (limit) of the population mean: $\bar{x} \pm \frac{s}{\sqrt{n}} \times t_{0.05}$

$$= 53 \pm \frac{3}{\sqrt{16}} \times 2.14 = 53 \pm \frac{3}{4} \times 2.14$$

$$= 54.5975 \ \& \ 51.4025$$

99% confidence limit of the population mean: $= \bar{x} \pm \frac{s}{\sqrt{n}} \times t_{0.01} = 53 \pm \frac{3}{\sqrt{16}} \times 2.95$

$$= 55.2125 \ \& \ 50.7875$$

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Q. The life time of electric bulbs for a random sample of 10 from a large consignment got the following data:

Item	1	2	3	4	5	6	7	8	9	10
life in ... 000 hr	4.2	4.6	3.9	4.1	5.2	3.8	3.9	4.3	4.4	5.6

Can we accept the hypothesis that the average lifetime of bulb is 4000 hrs?

⟹ Null Hypothesis; H_0: There is no significant difference in the sample mean and population mean. i.e; $\mu = 4000$ hrs.

Calculating mean of the sample,

$$\bar{x} = \frac{\Sigma x}{n} = \frac{4.2 + 4.6 + \cdots + 5.6}{10}$$

$$= \frac{44}{10} = 4.4$$

x	f	$x - \bar{x}$	$(x-\bar{x})^2$
4.2		−0.2	0.04
4.6		0.2	0.04
3.9		0.5	0.25
4.1		−0.3	0.09
5.2		0.8	0.64
3.8		−0.6	0.36
3.9		−0.5	0.25
4.3		−0.1	0.01
4.4		0	0
5.6		1.2	1.44

$\Sigma(x-\bar{x})^2 = 3.12$

$$s = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n-1}} = \sqrt{\frac{3.12}{9}} = 0.589$$

$$\cancel{t = \frac{\bar{x}-\mu}{s\sqrt{n}} = \frac{4.4 - 0.4 \times 10^3}{0.59 \times 10} = \frac{400}{0.59\sqrt{10}} = \frac{400\sqrt{10}}{0.59} = 214 = 2143.9}$$

$$t = \frac{\bar{x} - \mu}{s\sqrt{n}} = \frac{4.4 - 4}{\frac{0.589}{\sqrt{10}}} = 2.123$$

Degree of freedom = $n-1 = 10-1 = 9$

$t_{0.05} = 2.026$

Conclusion: Since, the calculated value of t is less than the tabulated value of t at 5% level of significance.

Level of confidence = $\bar{x} \pm \frac{s}{\sqrt{n}} \times t_{0.05}$

$= 4.4 \pm \frac{0.59}{\sqrt{10}} \times 2.026$

$= 4.4 \pm 0.422$

$+ \qquad -$

$4.822 \qquad 4.022$

∴ The null hypothesis $\mu = 4000$ hrs. is accepted. i.e. the average lifetime of bulbs could be 4000 hrs.

Assignment (1) The individuals are choosen at random from the normal population of students and their marks are found to be 63, 63, 66, 67, 68, 69, 70, 70, 71, 71. In the light of these details, discuss the mean marks of the population of student is 66.

(2) A random sample of a 10 boys and the IQs 70, 120, 110, 101, 88, 83, 95, 98, 107, 100

Do these data support the assumptions of the population mean 160?

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Ex- Prices of the shares of a company on different days in a month were found to be 66, 65, 69, 70, 69, 71, 70, 63, 64 and 68. Discuss whether the price of the share be 65. The table value of $t = 2.262$.

⇒ Let us take the hypothesis: There is no difference between the mean price of the shares and hypothetical population mean.

x	$x - \bar{x}$	$(x - \bar{x})^2$
66	-1.5	2.25
65	-2.5	6.25
69	1.5	2.25
70	2.5	6.25
69	1.5	2.25
71	3.5	12.25
70	2.5	6.25
63	-4.5	20.25
64	-3.5	12.25
68	0.5	0.25
		$\Sigma = 70.5$

Note: $S = \sqrt{\dfrac{\Sigma(X-\bar{X})^2}{n-1}}$

We know, $d = X - A$, $X = d + A$

$X - \bar{X} = d + A - \bar{X}$

$$S = \sqrt{\frac{\Sigma d^2 - n\Sigma \bar{d}^2}{n-1}}$$

$$\bar{X} = A + \frac{\Sigma d}{n}$$

$$X - \bar{X} = d + A - \left(A + \frac{\Sigma d}{n}\right) = d - \frac{\Sigma d}{n}$$

$$(X-\bar{X})^2 = \left(d - \frac{\Sigma d}{n}\right)^2$$

$$= d^2 + \frac{\Sigma d^2}{n^2} - \frac{2d\Sigma d}{n}$$

$$\Rightarrow \Sigma(X-\bar{X})^2 = \Sigma\left[d^2 + \frac{\Sigma d^2}{n^2} - \frac{2d\Sigma d}{n}\right]$$

$$= \Sigma d^2 - n\Sigma(\bar{d})^2$$

$$S = \sqrt{\frac{\Sigma d^2 - n(\Sigma \bar{d})^2}{n-1}}$$

$$S = \sqrt{\frac{\Sigma(X-\bar{X})^2}{n-1}} = \sqrt{\frac{70.5}{9}} = 2.799$$

By t-test, $t = \frac{\bar{x} - \mu}{S/\sqrt{n}} = \frac{67.5 - 65 \times \sqrt{10}}{2.799}$

$$= \frac{2.5 \times \sqrt{10}}{2.799} = 2.82$$

Degree of freedom $= n-1 = 10-1 = 9$

$t_{0.05} = 2.262$

The calculated value of t is greater than the tabulated value.
i.e hypothesis rejected
Hence, the mean prices of the share ~~is~~ could not be Rs. 65.

Q. Slim trim agency conducting a weight reduction program, claims that participants in their program achieve a weight reduction of atleast 5 kg after 2 weeks of the program. In evidence thereof, they ~~a~~ have given the following data on 10 participants who ha undergone this program, about their weights in kg. Prior to the program and two weeks after the program. On the basis of the sample evidence, can the claim of the agency on weight reduction be sustained?
Test at a significance level of 5%.
(Given $\nu = 9$, $t_{0.05} = 2.202$)

Before (kg): 86 92 100 93 88 80 88 92 95 106

After (kg): 77 84 92 87 80 74 80 85 95 96

but take difference 2nd − 1st.

⇒ Let us take the hypothesis that there is no significant difference in weight reduction before and after the program.

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S.No.	Before (kg)	After (kg)	d	d^2
1	86	77	9	81
2	92	84	8	64
3	100	92	8	64
4	93	87	6	36
5	88	80	8	64
6	80	74	6	36
7	88	80	8	64
8	92	85	7	49
9	95	95	0	0
10	106	96	10	100
			$\Sigma d = 70$	$\Sigma d^2 = 558$

$$\bar{d} = \frac{\Sigma d}{10} = \frac{70}{10} = 7$$

$$S = \sqrt{\frac{\Sigma d^2 - n(\bar{d})^2}{n-1}}$$

$$= \sqrt{\frac{558 - 10(\bar{d})^2}{n-1}} = \sqrt{\frac{558 - 490}{9}}$$

$$= \sqrt{\frac{68}{9}} = 2.749$$

Degree of freedom =

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}} = \frac{7}{2.749} \times \sqrt{10} = 2.3$$

$t =$

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Degree of the freedom $= n-1 = 9$

$t_{0.05} = \cancel{2.262}\ 2.28$

The calculated value of t is higher than the table value. The hypothesis is rejected. Hence, there is a significant difference b/w weight before & after the program.

Q. 12 students were given intensive coaching and 5 tests were conducted in a month. The scores of tests 1 and 5 are given below.

No. of students	1	2	3	4	5	6	7	8	9	10	11	12
Marks in 1st test	50	42	51	26	35	42	60	41	70	55	62	38
Marks in 5th test	62	40	61	35	30	52	68	51	84	63	72	50

Do the data indicate any improvement in the scores obtained in tests 1 and 5.

⇒ Let us take the hypothesis that there is no improvement in the scores obtained in the first and fifth tests, that is,

$H_0: \mu_d = 0$ and $H_1: \mu_d \neq 0$ (Two-tailed test)

No. of students	Marks in 1st Test	Marks in 5th test	Difference in Marks (d)	d^2
1	50	62	12	144
2	42	40	-2	4

3	51	61	+10	100
4	26	35	9	81
5	35	30	-5	25
6	42	52	10	100
7	60	68	8	64
8	41	51	10	100
9	70	84	14	196
10	55	63	8	64
11	62	72	10	100
12	38	50	12	144
			96	1122

$$\bar{d} = \frac{\Sigma d}{n} = \frac{\cancel{96}}{\cancel{12}} = 8$$

$$S_d = \sqrt{\frac{\Sigma d^2}{n-1} - \frac{(\Sigma d)^2}{n(n-1)}} = \sqrt{\frac{1122}{11} - \frac{(96)^2}{12 \times 11}}$$

$$= \sqrt{\frac{\overset{102}{\cancel{1122}}}{\cancel{11}} - \frac{96 \times \cancel{96}\,8}{\cancel{12} \times 11}} = \sqrt{32.181} = 5.673$$

Applying t-statistic, we have

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}} = \frac{\bar{d} - \mu_d}{S_d/\sqrt{n}} = \frac{8-0}{5.673/\sqrt{12}}$$

$$= \frac{8 \times 3.4641}{5.673} = 4.885$$

Since the calculated value $t_{cal} = 4.885$ is more than its critical value, $t_{0.05}$ at degree of freedom 11 i.e. 2.20, the null hypothesis is rejected. Hence, we conclude there is an improvement in the scores obtained in two tests.

Q. To verify whether a course in accounting improved performance, a similar test was given to 12 participants both before and after the course. The original marks recorded in alphabetical order of the participants were 44, 40, 61, 52, 32, 44, 70, 41, 67, 72, 53 and 72. After the course, the marks were in the same order: 53, 38, 69, 57, 46, 39, 73, 48, 73, 74, 60, and 78. Test whether the course was useful?

⇒ Let us take the null hypothesis that the course has not improved the performance of the participants. Applying t-test,

Before	After	2nd - Ist d	d^2
44	53	9	81
40	38	-2	4
61	69	8	64
52	57	5	25
32	46	14	196
44	39	-5	25
70	73	3	9
41	48	7	49
67	73	6	36
72	74	2	4

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53	60	7	49
72	78	6	36
		$\Sigma d = 60$	$\Sigma d^2 = 578$

$$\bar{d} = \frac{\Sigma d}{n} = \frac{60}{12} = 5$$

$$s = \sqrt{\frac{\Sigma d^2 - n(\bar{d})^2}{n-1}} = \sqrt{\frac{578 - 12(5)^2}{11}} = \sqrt{\frac{578-300}{11}}$$

$$= \sqrt{\frac{278}{11}} = \sqrt{25.27} = 5.027$$

$$t = \frac{\bar{d}\sqrt{n}}{s} = \frac{5\sqrt{12}}{5.027} = \frac{17.32}{5.027} = 3.45$$

For degree of freedom $= 11$, $t_{0.05} = 2.201$. The calculated value of t is more than the table value of t. The null hypothesis is rejected. Hence, course has improved the performance of the participants.

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When we have two different mean (and two different standard deviation).

If $x_1, x_2, \dots, x_{n_1}$ and $y_1, y_2, \dots, y_{n_2}$ are two different samples of sizes n_1 and n_2. It's drawn from the two normal population, with mean μ_1 and μ_2, then by t-test, $t = \dfrac{\bar{x} - \bar{y}}{S\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$

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Note: ① If the two samples standard deviation S_1, S_2 are given. Then, we have

$$S^2 = \frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2 - 2}$$

② If S_1, S_2 are not given, then ~~$S^2 = \Sigma(x$~~

$$S^2 = \frac{\Sigma(x_1 - \bar{x}_1)^2 + \Sigma(y_1 - \bar{y}_1)^2}{n_1 + n_2 - 2}$$

$$S^2 = \frac{\Sigma(x_1 - \bar{x}_1)^2 + \Sigma(x_2 - \bar{x}_2)^2}{n_1 + n_2 - 2}$$

~~$S^2 \neq$~~

Q. Two samples of a sodium vapour bulbs were tested for length of life and the following results were got:

	Size	Sample mean	Sample S.D.
Type I	8	1234 hrs.	36 hrs.
Type II	7	1036 hrs.	40 hrs.

Is the difference in the mean significant to generalize that type I is superior to type II regarding length of life?

⇒ Null Hypothesis: $H_0: \mu_1 = \mu_2$ i.e. two types of bulbs have same lifetime and $H_1: \mu_1 > \mu_2$ i.e., type I is superior to type II.

According to the question, we have $n_1 = 8$, $n_2 = 7$, $S_1 (SD) = 36$, $S_2 (SD) = 40$,

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$$S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

$$S^2 = \frac{(36)^2 \times 8 + (40)^2 \times 7}{8+7-2}$$

$$= \frac{1296 \times 8 + 1600 \times 7}{13} = 1659.076$$

$$S = \sqrt{1659.076} = 40.7317$$

By using t-test formula,

$$t = \frac{\bar{x} - \bar{y}}{S\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{1234 - 1036}{40.7317\sqrt{\frac{1}{\cancel{36}8} + \frac{1}{7}}}$$

$$= \frac{198}{40.7317\sqrt{\frac{15}{56}}}$$

$$= \frac{198}{40.7317 \times 0.5175}$$

$$t = 9.39 = 18.1480$$

$$df = n_1 + n_2 - 2 = 8 + 7 - 2 = 13$$

Level of significance $t_{0.05} = 1.77$

$|t| > t_{\text{tabulated value at 5\% level of significance}}$

$18.1480 > 1.77$

Hence, hypothesis rejected i.e. type I is

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definitely superior to type II.

Q Samples of a sizes 10 and 14 were taken from two normal populations with standard deviation 3·5 and 5·2. The samples mean were found to be 20·3 and 18·6. Test whether the means of a the two populations are the same at 5% level.

⇒ Null Hypothesis $H_0 : \mu_1 = \mu_2$ Means of two population are same.

Alternative Hypothesis: $H_1 : \mu_1 \neq \mu_2$

According to the question, $n_1 = 10$, $n_2 = 14$, $s_1 = 3·5$, $s_2 = 5·2$

$$S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2} = \frac{10 \times 12·25 + 14 \times (5·2)^2}{24 - 2}$$

$$= \frac{122·5 + 378·56}{2}$$

$$= 22·775$$

$$S = \sqrt{S^2} = 4·772.$$

$$t = \frac{\bar{x} - \bar{y}}{S\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{20·3 - 18·6}{4·772\sqrt{\frac{14+10}{140}}}.$$

$$= \frac{1·7}{4·772\sqrt{24/140}} \qquad = \frac{1·7}{1·9} = 0·8604.$$

Degree of freedom $= 22$

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Level of significance, $t_{0.05} = 2.0739$

Conclusion: Since, $|t| < t_{0.05}$ (tabulated value) $0.8604 < 2.0739$.
Hypothesis accepted i.e. there is no significant difference b/w their mean.

Assignment: The height of a 6 randomly chosen sailors in ~~sello~~ inches are 63, 65, 68, 69, 71, 72. Those of 9 randomly chosen soldiers are 61, 62, 65, 66, 69, 70, 71, 72, 73. Test whether the sailors are on the average taller than soldiers. Level of significance $t_0 = 1.77$

(χ^2 distribution)
(Chi Squared Distribution)

The χ^2 test was first used by Karl Pearson in the year 1900.
The quantity χ^2 describes the magnitude of the difference between observation and theory.
It is denoted by χ^2 and defined as

$$\chi^2 = \Sigma\left(\frac{(O_i - E_i)^2}{E_i}\right) \quad \text{or} \quad \chi^2 = \Sigma \frac{(O-E)^2}{E}$$

where O refers to the observed frequencies and E refers to the expected frequency

Note: In general, the expected frequency for any call can be calculated from the following equation:

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$$\bar{E} = \frac{RT \times CT}{N}$$

where $\bar{E}$ = expected frequency, RT = the row total for the row containing the cell.
CT = the column total for the column containing the cell

Degree of freedom: To calculate the degree of freedom of χ^2 test : $n-1$
$(R-1)(C-1)$

Ex- In an anti-malarian ~~in a~~ ~~certi~~ campaign in a certain area quinine was administrated to 812 persons. Out of a total ~~300~~ 3248, the no. of fever cases is shown below:

Treatment	Fever	No Fever	Total
Quinine	20	792	812
No-Quinine	220	2216	2436
Total	240	3008	3248

Discuss the usefulness of quinine in checking malaria.

⇒ Hypothesis: Let us take the hypothesis that the quinine is not effective in checking malaria.

$$E(AB) = \frac{A \times B}{N} = \frac{RT \times CT}{N}$$

$$= \frac{\cancel{240} \times \cancel{812}}{\cancel{3248}} = 60$$

The table of expected frequencies shall

Treatment	Fever	No-Fever	Total
Quinine	60	752	812
No-quinine	180	2256	2436
Total	240	3008	3248

O	E	O − E	$(O-E)^2$
20	60	−40	1600
220	180	40	1600
792	752	40	1600
2216	2256	−40	1600

$$\frac{1600}{3248} = 0.492$$

$26.66 + 8.88 + 2.12 + 0.709$

$= 38.36$

By using χ^2 test, $\chi^2 = \sum \frac{(O-E)^2}{E}$

$= 38.393$

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calculating degree of freedom $= (R-1)(C-1)$
$= (2-1)(2-1) = 1$

Level of significance, $\chi^2_{0.05} = 3.84$.

~~The~~

Conclusion: The calculated value of the χ^2 is greater than the table value. The hypothesis is rejected, hence quinine is useful in checking malaria.

Q. Based on information on 1000 randomly selected fields about the tenancy status of the cultivators of these fields and use of fertilizers, collected in an agroeconomic survey, the following classification was noted.

	Owned	Rented	Total
Using fertilizers	416	184	600
Not using fertilizers	64	336	400
Total	480	520	1000

Would you conclude that the owner-cultivators are more inclined towards the use of fertilizers at a 5% level of significance.
Carryout a χ^2 test as per testing procedure.

Sol: Let us take the hypothesis that ownership of fields and the use of fertilizers are independent attributes.

$$Exp(AB) = \frac{A \times B}{N} = \frac{48\cancel{0} \times 6\cancel{00}}{\cancel{1000}} = 288$$

	288	312	600
	192	208	400
Total	480	520	1000

O	E	O−E	$(O-E)^2$	$\frac{(O-E)^2}{E}$
416	288	128	16384	56.889
64	192	−128	16384	85.333
184	312	−128	16384	52.513
336	208	128	16384	78.769
				$\Sigma = 273.534$

By using χ^2 test, $\chi^2 = \frac{\Sigma (O-E)^2}{E}$

$= 273.534$

calculating degree of freedom $= (R-1)(C-1)$

$= (2-1)(2-1)$

$= 1$

Level of significance $\chi_{0.05} = 3.84$

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Conclusion: The calculated value of χ^2 is much more than the table value. The hypothesis H_0 is rejected. Hence, it can be concluded that the owner cultivators are more inclined towards the use of fertilizers.

Q In an experiment on ~~humini~~ immunization of cattles from tuberclosis ~~some disease~~, the following results were obtained ~~inoculated~~

	Affected	Not Affected
Inoculated	12	26
Not-Inoculated	16	6

~~Total~~

Calculate χ^2 and discuss the effect of vaccine in controlling susceptibility to tuberclosis ~~this inoculated~~ (5% ~~level~~ value of χ^2 for one degree of freedom = 3.84)

Expectation of AB

$\Rightarrow$ $E(AB) = \frac{\cancel{38}\ 28 \times \cancel{32}\ 38}{60} = 17.734$

Expected value table E

17.7	20.3	38
10.3	11.7	22
28	32	60

Hypothesis: Vaccine is not effective in controlling the susceptility to tuberclosis.

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O	E	(O−E)	$(O-E)^2$	$\frac{(O-E)^2}{E}$
12	17.7	−5.7	32.49	1.836
16	10.3	5.7	32.49	3.154
26	20.3	5.7	32.49	1.600
6	11.7	−5.7	32.49	2.777

By using χ^2 test, $\sum \frac{(O-E)^2}{E} = 9.365$

Degree of freedom $= (r-1)(c-1)$
$= (2-1)(2-1) = 1$

Level of significance, $\chi^2_{0.05} = 3.84$

Conclusion: Since, the calculated value of χ^2 is greater than the table value, the hypothesis is not true.

∴, we conclude that vaccine is effective in controlling susceptibility to tuberclosis.